

07-Memory-Architecture.md

Healthcare Clinical Intelligence Platform

Enterprise Memory Architecture

Version: 1.0

Status: Approved Design

Mission

Build a governed memory system that:

- Personalizes responses
 - Preserves organizational knowledge
 - Supports long-term context
 - Maintains compliance
 - Prevents memory corruption
 - Enables future agentic capabilities
-

Memory Philosophy

Bad:

Conversation
↓
Forget Everything

Good:

Conversation
↓
Memory Extraction
↓
Governance
↓
Memory Store

↓

Future Retrieval

Memory Layers

Layer 1

Session Memory

Layer 2

User Memory

Layer 3

Department Memory

Layer 4

Organization Memory

Layer 5

Agent Memory

Memory Hierarchy

Organization Memory

|

Department Memory

|

User Memory

|

Session Memory

Session Memory

Purpose:

Current conversation context.

Storage

Supabase

Redis Cache

Lifetime

Current Session

Configurable TTL

Examples

Recent Questions

Current Topic

Current Documents

Current Citations

User Memory

Purpose:

Personalization.

Examples

Medical Specialty

Preferred Response Style

Frequently Used Guidelines

Research Interests

Commonly Accessed KBs

Example

```
{  
  "specialty": "cardiology",  
  "preferred_response": "detailed",  
  "research_focus": "heart_failure"  
}
```

User Memory Types

Preferences

Behavioral

Interaction

Professional

Research

Department Memory

Purpose:

Shared department intelligence.

Examples

Cardiology Protocols

Department Decisions

Department Knowledge Assets

Department Preferences

Access

Department Members Only

Organization Memory

Purpose:

Hospital-wide intelligence.

Examples

Policies

Treatment Standards

Organization Guidelines

Institutional Preferences

Access

Organization Controlled

Agent Memory

Purpose:

Improve future agent behavior.

Examples

Failed Retrieval Patterns

Successful Retrieval Patterns

Prompt Improvements

Workflow Improvements

Governance Rule

Agent Memory cannot become production knowledge automatically.

Flow

```
graph TD; A[Agent Reflection] --> B[Memory Candidate]; B --> C[Review Queue]; C --> D[Approval]; D --> E[Memory Store];
```

Memory Storage Architecture

PostgreSQL

Stores:

Structured Memory

Metadata

Ownership

Governance Status

Qdrant

Stores:

Memory Embeddings

Semantic Memory Search

Redis

Stores:

Hot Memory

Recent Context

Memory Schema

User Memory

```
{
  "memory_type": "preference",
  "importance": 0.95,
  "content": "Prefers detailed responses"
}
```

Department Memory

```
{
  "memory_type": "department_policy",
  "department": "cardiology"
}
```

Organization Memory

```
{  
  "memory_type": "guideline",  
  "organization": "hospital"  
}
```

Memory Lifecycle

Create

↓

Review

↓

Store

↓

Retrieve

↓

Update

↓

Archive

Memory Creation

Sources:

Conversation

Feedback

Document Activity

Agent Reflection

Admin Input

Memory Extraction Pipeline

```
graph TD; Conversation --> ME[Memory Extractor]; ME --> IS[Importance Scoring]; IS --> GC[Governance Check]; GC --> MS[Memory Store];
```

Importance Scoring

Scale:

0.0 → 1.0

Factors

Frequency

Recency

User Feedback

Business Value

Medical Relevance

Memory Retrieval

Before every response:

```
graph TD;
    A[User Query] --> B[Relevant Memory Search];
    B --> C[Context Assembly];
    C --> D[Agent Processing];
```

Memory Search Strategy

Metadata Filter

↓

Semantic Search

↓

Recency Ranking

↓

Importance Ranking

Memory Limits

Prevent unlimited growth.

User Memory

Top Important Memories

Archive low-value memories.

Department Memory

Periodic consolidation.

Organization Memory

Version controlled.

Memory Compression

Old memories:

100 Memories

|

Summarization

|

10 High-Level Memories

Memory Governance

Every memory contains:

Organization

Department

Owner

Source

Created By

Approval Status

Version

Memory Security

Memory must obey:

RBAC

Department Boundaries

Tenant Isolation

Audit Requirements

Memory Audit

Track:

Created

Updated

Retrieved

Deleted

Archived

Memory Observability

Metrics:

Memory Count

Memory Usage

Memory Hit Rate

Memory Recall

Memory Latency

Memory Quality

Memory Evaluation

Measure:

Relevance

Usefulness

Accuracy

Impact On Answer Quality

Long-Term Learning

Allowed:

Preferences

Workflow Optimizations

Retrieval Improvements

Not Allowed:

Medical Knowledge Changes

Policy Changes

Guideline Changes

Without Approval

Memory Retention

User Memory

Configurable

Department Memory

Long-Term

Organization Memory

Permanent

Versioned

Future Enhancements

Temporal Memory

Knowledge Evolution Tracking

Department Intelligence Graphs

Memory Recommendation Engine

Cross-Agent Shared Memory

Final Principle

Memory exists to improve context.

Memory must never bypass governance.

Trust and compliance are more important than remembering everything.